

Mahima Pal

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Summary

Computer Science (AI/ML) undergraduate with hands-on experience architecting full-stack, multi-agent AI systems, explainable decision-support engines, and RAG pipelines using LangGraph and FastAPI, shipping production-grade APIs at sub-200ms latency. Smart India Hackathon 2023 Top 5 finalist among 225 teams.

Education

B.Tech, Computer Science Engineering, Vellore Institute of Technology Bhopal
CGPA: 8.3/10

2023 – 2027

Technical Skills

- **AI/ML Frameworks:** LangGraph, LangChain, TensorFlow, Keras, Scikit-learn, NumPy, Pandas
- **Deployment & Infra:** FastAPI, REST APIs, SSE, Docker, Git
- **Concepts:** RAG Pipelines, Multi-Agent Systems, Explainable AI, Prompt Engineering, Model Optimization, Data Visualization, Agentic AI
- **Languages:** Python, SQL

Work Experience

Larsen & Toubro (L&T)

Data Analysis & Management Intern

Guwahati, Assam

May 2025 – Jun 2025

- Streamlined data collection, validation, and reporting processes across internal data analysis and management initiatives to support informed decision-making.
- Analyzed large volumes of operational and project data to identify trends, inconsistencies, and improvement opportunities, enhancing data accuracy and reliability.
- Partnered with cross-functional teams to gather and validate requirements.
- Co-developed standardized data management protocols, improving consistency and quality across departments.

Projects

ClarityDoc AI – Multi-Agent Document Intelligence Platform [[GitHub](#)]

Apr 2026 – Jul 2026

FastAPI, React, Vite, Google GenAI SDK (Gemini-2.5-Flash), Pydantic, pypdf, python-docx, Render

- Engineered a full-stack app (FastAPI + React/Vite) deployed on Render as a unified service, achieving sub-200ms API latencies.
- Developed a multi-agent routing engine (Gemini-2.5-Flash + Pydantic) classifying documents into 4 domains with 98%+ accuracy, routing to tailored agents.
- Implemented PDF/DOCX/TXT extraction pipelines with a 12,000-char input sampler, cutting LLM token usage by 65%.
- Designed schema-driven extraction with an interactive UI (risk indicators, flash-cards, quizzes), reducing time-to-insight by 80%.

CitySurvi – City Affordability Planner & Relocation Decision Engine

2026

Python, FastAPI, Pydantic, Docker, Pytest, Vanilla JS (ES6+), HTML5/CSS3

- Engineered an Explainable AI (XAI) decision engine analyzing multi-dimensional financial profiles across 10 tier-1 Indian tech cities, generating relocation verdicts with 0% hallucination risk by grounding logic on deterministic mathematical baselines.
- Designed a 5-factor weighted financial stress scoring algorithm (0–100 index) – rent burden, savings deficit, essential/discretionary spend, cashflow buffer – classifying users into 5 risk bands.
- Built an asynchronous "What-If" scenario simulation engine running real-time multi-variable sensitivity analysis with sub-50ms latency, projecting 1–3 year cumulative wealth multipliers, plus a PPP relocation evaluator computing break-even CTC adjustments across major Indian IT corridors.
- Built a Dockerized REST API with Pydantic data contracts and automated unit/regression test suites, achieving <100ms response times across all endpoints.

Daily Brief AI – Real-Time Agentic News & Weather Assistant [[GitHub](#)]

Dec 2025 – Jul 2026

FastAPI, SSE, Groq SDK (Llama-3.1-8b), React, Vite, Google OAuth 2.0, APScheduler

- Constructed a FastAPI + SSE backend streaming agent logs, tool traces, and briefs, cutting latency to under 250ms.
- Assembled an ingestion pipeline over Google News RSS and OpenWeather/Open-Meteo APIs, deduplicating and ranking 50+ articles into a top-20 list in under 1.5s.
- Implemented Google OAuth 2.0 for Gmail access, automating daily brief generation and delivery via APScheduler; delivered a React + Vite SPA consuming chunked streaming responses, reducing load time by 30%.

Achievements

- **Smart India Hackathon 2023:** Top 5 Finalist among 225 competing teams.

Certifications

Bits & Bytes in Computer Networking (Google) | NPTEL Foundations of Machine Learning (SWAYAM) | NPTEL Marketing Analytics (SWAYAM) | Generative AI (Oracle)